SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH109
Calculus

Tutor-Marked Assignment

July 2025 Presentation

TUTOR-MARKED ASSIGNMENT (TMA)

This assignment is worth 16% of the final mark for MTH109 Calculus.

The cut-off date for this assignment is **21 October 2025, 2355hrs.**

Note to Students:

You are to include the following particulars in your submission: Course Code, Title of the TMA, SUSS PI No., Your Name, and Submission Date.

For example, ABC123_TMA01_Sally001_TanMeiMeiSally (omit D/O, S/O). Use underscore and not space.

Question 1

Determine the following limits.

(a) $$\lim_{x\to 3} \frac{3x^2-2x+1}{\sqrt[3]{x^2+5x+3}}$$

(3 marks)

(b) $$\lim_{x\to 6} \frac{4-\sqrt{x+10}}{6-x}$$

(3 marks)

(c) $$\lim_{x\to 1} \frac{\sqrt[4]{x}-1}{x-1}$$

(3 marks)

(d) $$\lim_{x\to 0} x^{25}\sin\left(\frac{2}{x}\right)$$

(3 marks)

(e) $$\lim_{x\to 0} (\csc(x))^{x^5}$$

(3 marks)

Question 2

Stone skipping is the ancient art of casting a flat stone across water so that it bounces off the surface. The current world record for the most number of consecutive skips of a stone in water is 88, achieved by Kurt Steiner in 2013 (https://www.guinnessworldrecords.com/world-records/most-skips-of-a-skimming-stone). For a demonstration, please see https://www.youtube.com/watch?v=bfvsPtAEvOs [New China TV (2023, 1 February). *Chinese boy showcases super skill of stone skipping.* [Video]. YouTube.]

This problem considers a simplified model for stone skipping. Consider a flat square stone with negligible thickness and of mass M. The stone is thrown over a flat water surface such that its velocity is V and the angle between the stone surface and the water plane, called the **tilt angle**, is θ. The angle between the velocity V and the horizontal, called the **incidence angle**, is β. The component of V parallel to the water surface is denoted by V_X and the component of V perpendicular to the water surface is denoted by V_Z. Figure Q2 gives a schematic view of the process during which the stone is partially immersed in water. The immersed area is A_{im} and the depth of the immersed edge of the stone is z. This model uses the convention that $z \leq 0$. Furthermore, it is assumed that the tilt angle θ and the incidence angle β are constant. Time $t = 0$ corresponds to the instant when the edge of the stone reaches the water surface. The initial components of the velocity are V_{x_0} and V_{z_0} ($V_{z_0} \leq 0$), parallel and perpendicular to the water surface, respectively. The velocity V is approximated as $V^2 \approx V_{x_0}^2$.

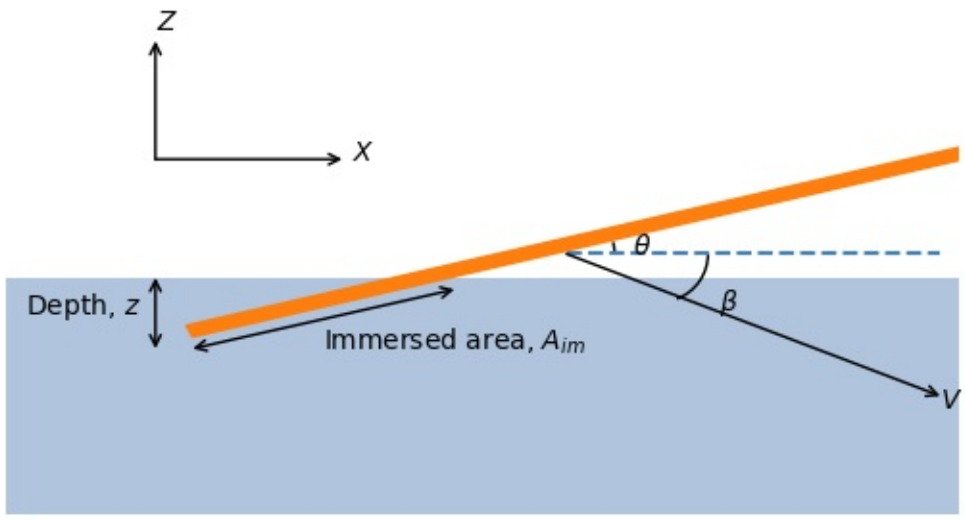

Figure Q2

The equations of motion for the collisional process are

$$M\frac{dV_X}{dt} = -\frac{1}{2}\rho_w V_{x_0}^2 A_{im}\left(C_\ell \sin(\theta) + C_f \cos(\theta)\right) \tag{1}$$

$$M\frac{dV_Z}{dt} = -Mg + \frac{1}{2}\rho_w V_{x_0}^2 A_{im}\left(C_\ell \cos(\theta) - C_f \sin(\theta)\right) \tag{2}$$

Here g is the acceleration due to gravity, C_ℓ and C_f are the lift and friction coefficients (assumed to be constant) and ρ_w is the mass density of water. The length of one edge of the stone is a. Then, the immersed area A_{im} equals $\frac{a|z|}{\sin(\theta)} = -\frac{az}{\sin(\theta)}$ (since $z \leq 0$). Define

$$C = C_\ell \cos(\theta) - C_f \sin(\theta).$$

As $V_Z = \frac{dz}{dt}$, Equation (2) reduces to

$$\frac{d^2z}{dt^2} + \omega_0^2 z = -g, \tag{3}$$

where ω_0 is the **characteristic frequency** defined by

$$\omega_0^2 = \frac{C\rho_w V_{x_0}^2 a}{2M\sin(\theta)}.$$

(a) Show that $z(t) = -\frac{g}{\omega_0^2} + \frac{g}{\omega_0^2}\cos(\omega_0 t) + \frac{V_{z_0}}{\omega_0}\sin(\omega_0 t)$ satisfies Equation (3) and at $t = 0$, $z(t) = 0$ and $\frac{dz(t)}{dt} = V_{z_0}$.

(5 marks)

(b) Using the values $M = 0.1$ kg, $C_\ell = C_f = 1$, $a = 0.1$ m, $g = 10$ ms^{-2}, $\rho_w = 1000$ kgm^{-3}, $\theta = \beta = 15^\circ$ and $V_{x_0} = 1$ ms^{-1}, apply derivative tests to calculate the absolute maximum value of $|z(t)|$ on $\left[0, \frac{2\pi}{\omega_0}\right]$. (This is equal to the largest magnitude immersed depth attained by the stone before it emerges totally from the water surface; remember that $z \leq 0$.) Give your answer correct to 3 decimal places.

(10 marks)

Question 3

(a) Solve each of the following derivatives.

(i) $$\frac{d}{dx}\int_0^x \frac{9-y^2}{1+y^3}\,dy$$

(5 marks)

(ii) $$\frac{d}{dx}\int_{\sqrt{x}}^1 \sin^3(\cos(t))\,dt$$

(5 marks)

(iii) $$\frac{d}{dx}\int_{\cos(x)}^{\sin(x)} \ln(t^4+1)\,dt$$

(5 marks)

(b) Determine the values of a and b such that the function $f(x) = ax + b$ satisfies $f(1) = 1$ and $\int_0^1 |f(x)|\,dx$ has the minimum possible value.

(5 marks)

Question 4

Solar energy is currently Singapore's most prevalent renewable energy source. Installed solar capacity saw a more than thousand-fold increase in 15 years, from 0.0004 gigawatt-peak (GWp) in 2008 to 1.17 GWp in 2023. The Solar Generation Profile (SGP) is a model used to estimate solar generation output in Singapore and is based on factors such as Singapore's historical average solar irradiance from 7am to 7pm.

This problem considers two mathematical models to approximate the SGP. The first model consists of a single sinusoidal wave. In order to account for the variability of solar radiation due to weather changes, the second model is a superposition of multiple sinusoidal waves with different amplitudes and frequencies.

The SGP for an installed solar capacity of 1 kMWac over a 24-hour period is shown in Figure Q4. (MWac is a measure of power output from solar installations after the output of the photovoltaic (PV) panels have been converted to alternating current via inverter devices.)

In this question, give all numerical answers correct to 3 decimal places.

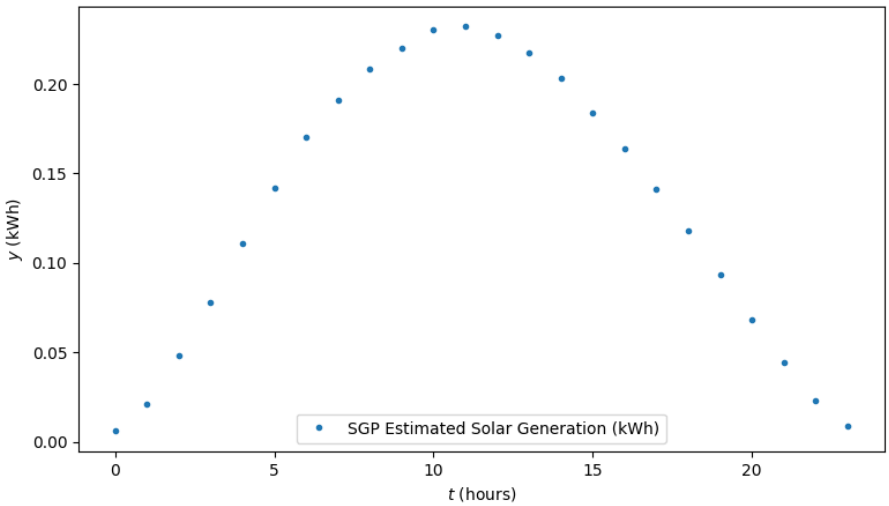

Figure Q4: SGP for an installed solar capacity of 1 kMWac over 24 hours [Data source: *Solar Generation Profile (effective from 1 December 2024)*, accessed on 13 April 2025, {https://www.ema.gov.sg/consumer-information/solar/solar-generation-profile}]

(a) In the first model, the estimated solar generation output in kWh, $y_1(t)$, is related to time in hours, t, by the equation

$$y_1(t) = 0.13117 - 0.10806\cos\left(\frac{2\pi(t+0.90831)}{24}\right). \tag{4}$$

Figure Q4a shows the graph of estimated solar generation output $y_1(t)$ (kWh) versus time t (hours) for the first model. Formulate a suitable integral based on Equation (4) and use it to estimate the average solar generation output (kWh) of a solar PV system with an installed capacity of 1 kMWac between $t = 0$ and $t = 23$.

(5 marks)

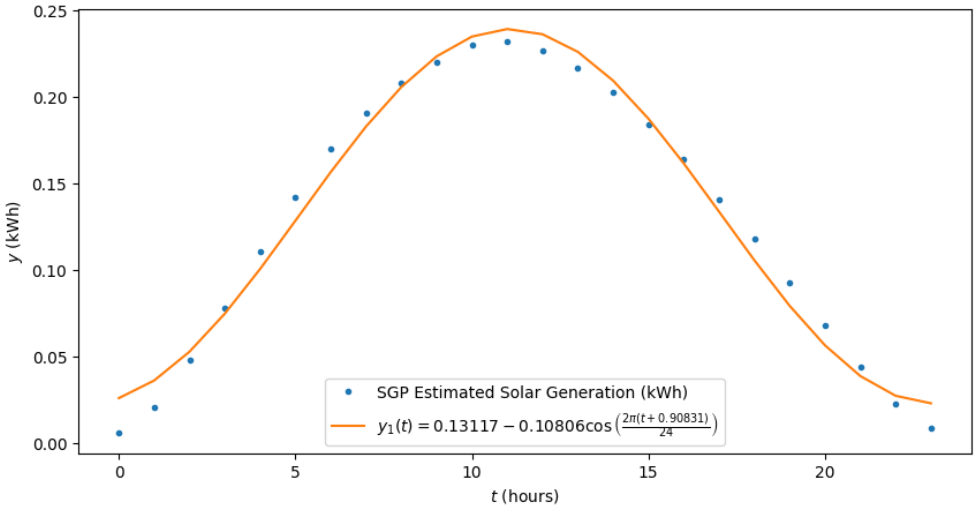

Figure Q4a: Single sinusoidal wave model

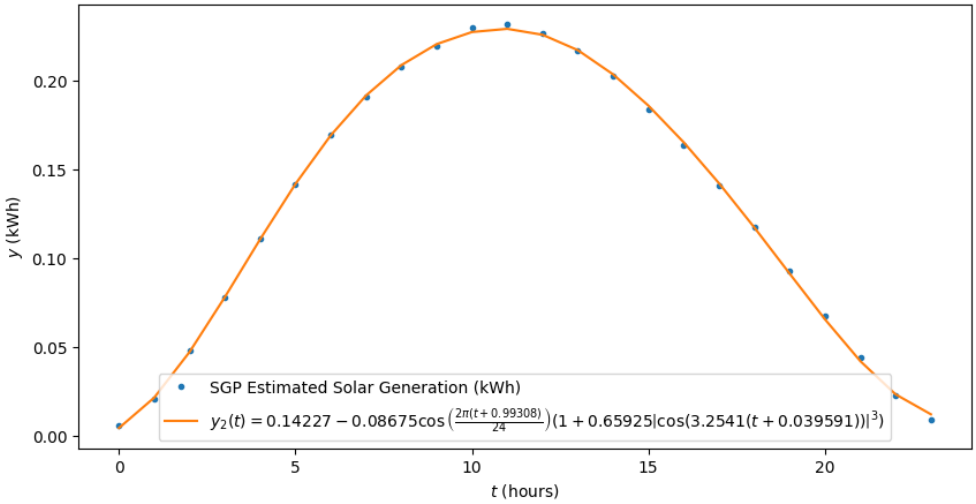

Figure Q4b: Multiple sinusoidal waves model

(b) In the second model, the estimated solar generation output in kWh, $y_2(t)$, is related to time in hours, t, by the equation

$$y_2(t) = 0.14227 - 0.08675\cos\left(\frac{2\pi(t+0.99308)}{24}\right)\left(1+0.65925\,|\cos(3.2541(t+0.039591))|^3\right). \tag{5}$$

Figure Q4b shows the graph of estimated solar generation output $y_2(t)$ (kWh) versus time t (hours) for the second model.

(i) Determine the set of points in $[0,23]$ where $y_2(t)$ is continuous. Explain your answer. (5 marks)

(ii) Determine the set of points in $(0,23)$ where $y_2(t)$ is differentiable. Explain your answer. (5 marks)

(iii) Calculate the equation of the tangent to the graph of $y_2(t)$ at $t = 11$. (5 marks)

Question 5

Employ the technique of partial fractions to evaluate the following integrals.

(a) $$\int \frac{4x}{x^2+2x-3}\,dx$$ (3 marks)

(b) $$\int \frac{8(2x+3)}{(x^2+4)(2-x)}\,dx$$ (3 marks)

(c) $$\int \frac{4}{(x-1)^2(3-x)}\,dx$$ (4 marks)

Question 6

Employ the technique of substitution to evaluate the following integrals.

(a) $$\int \frac{x^5}{\sqrt{1-x^3}}\,dx$$ (3 marks)

(b) $$\int x\cos(x^2)\sin(x^2)\,dx$$ (3 marks)

(c) $$\int \sqrt{x^6-x^8}\,dx$$ (4 marks)

Question 7

Employ the technique of integration by parts to evaluate the following integrals.

(a) $$\int x\sin(2x)\,dx$$ (3 marks)

(b) $$\int x\sin^2(x)\,dx$$ (3 marks)

(c) $$\int x\sin(x)\cos(3x)\,dx$$ (4 marks)

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